EXAM REVISION 1

MATHEMATICS PAPER 2 – GRADE 10

- This revision paper comprises 25 pages with 13 questions comprising many sub-questions.
- As many key concepts have been covered as possible.
- Grade 11 concepts sometimes covered in Grade 10 are not included.
- Answer all questions if possible.
- Guideline marks are provided for each question.
- Do not be concerned with the high total number of marks in some questions/sub-questions. This just facilitates more mathematical concepts being included for practise.
- Number your answers exactly as described in the questions.
- Clearly show all workings and calculations, as applicable.
- Correct answers only will not necessarily generate full marks.
- A non-programmable calculator may be used.
- All answers should be rounded to TWO decimal places unless otherwise directed.
- Diagrams are not necessarily drawn to scale.

NOTE: A Memorandum may be provided on request. It is certainly in your interest to attempt and apply your mind to all questions prior to receiving or making use of the memorandum Use self-marking to reinforce understanding of all concepts covered.

Think twice, write once, but most of all, ENJOY!

All the very best in all your school exams!

NB: Every effort has been made to present accurate questions and solutions, however please note that no third party checking has been applied whatsoever.

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DATA HANDLING

QUESTION 1

1.1 The following Stem and Leaf Plot represents single long jump distances to the nearest 0,1metre by athletes of various ability:

STEM	LEAF
5	2 ; 6 ; 7 ; 7
6	0 ; 3 ; 3 ; 5 ; 8
7	1 ; 2 ; 4 ; 7
8	0 ; 1 ; 3 ; 5
KEY: 5 | 2 = 5,2 metres	

1.1.1 How many jumpers took part? (1)

1.1.2 Write down the Mode. (2)

1.1.3 Determine the Mean (1 decimal place). (2)

1.1.4 Determine the Median. (2)

1.1.5 Determine the IQR. (3)

1.1.6 Hence plot a Box and Whisker Diagram. (4)

1.1.7 What percentage jumped more than 6,8m? (1)

1.2 A survey was done on what percentage of their salary 100 home owners paid in municipal service fees. The results are tabulated below:

% Salary	Frequency
(0 ; 5]	5
(5 ; 10]	15
(10 ; 15]	30
(15 ; 20]	40
(20 ; 25]	10

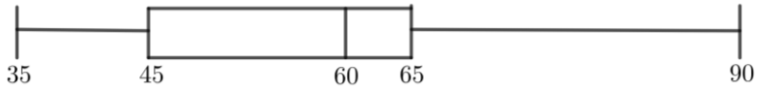
1.2.1 Draw a Histogram representing this data. (4)

1.2.2 In which class/group doe the 1st quartile lie? (2)

1.2.3 Calculate the average percentage of salary spent on municipal services. (3)

1.2.4 Explain why this average is an approximation. (1)

1.3 A Box and Whisker Diagram shows the results of a Grade 10 maths examination out of 100.



1.3.1 Write down the Range. (1)

1.3.2 Determine the Semi-Interquartile Range. (2)

1.3.3 What percentage of the pupils scored over 65% ? (2)

1.3.4 If 125 pupils scored between 35% and 60%, how many wrote the exam? (2)

QUESTION 2

2.1 The speed of 1000 vehicles was measured at a particular point in a street. The table below reflects the data

SPEED (km/h)	FREQUENCY
$0 < S \leq 20$	5
$20 < S \leq 40$	45
$40 < S \leq 60$	x
$60 < S \leq 80$	398
$80 < S \leq 100$	219
$100 < S \leq 120$	30
$120 < S \leq 140$	3

2.1.1 Determine the value of x. (2)

2.1.2 Write down the Modal Class. (1)

2.1.3 In which class/group does the 3rd quartile lie? (1)

2.1.4 If the speed limit was 80 km/h, how many drivers exceeded the speed limit? (2)

2.1.5 Calculate the approximate mean speed, giving your answer to the nearest whole number. (3)

2.2 The Histogram below shows the results of a survey measuring the number of shoppers visiting a shopping centre each week. Frequencies are indicated in brackets.



2.2.1 Write down the range of the highest number of shoppers during any week. (1)

2.2.2 Calculate the estimated total number of shoppers over the survey period. (3)

2.2.3 For how many weeks was the survey carried out? (2)

2.2.4 Hence determine an estimation of the mean, giving your answer to the nearest ten shoppers. (2)

2.2.5 Determine the estimated median. (3)

2.2.6 Determine in which class the 1st quartile lies. (2)

2.3 The marks of 14 learners in a mathematics test out of 100 are shown in the table below:

62	44	48	90	70	76	56
40	50	60	66	52	54	70

Determine the Five Number Summary and hence draw a neat Box and Whisker Diagram. Clearly show the positions of the Five Number Summary. Use a scale of $2mm = 1\ mark$. Thus the length from lowest to highest will be $50 \times 2mm = 100mm$. (6)

2.4 The following table consists of an ordered set of 7 values:

39	a	b	52	c	72	d

Determine the values of $a, b, c,$ and d if:

- The Mode is 72 and $c < d$.
- The IQR is 31
- $c = b + 16$
- The Mean is 56 (5)

2.5 An ordered list of positive integers is given in the table below. The mode is 15.

a	b	10	c	d	15	e	19

Determine the maximum possible sum $a + b + c + d + e$. (4)

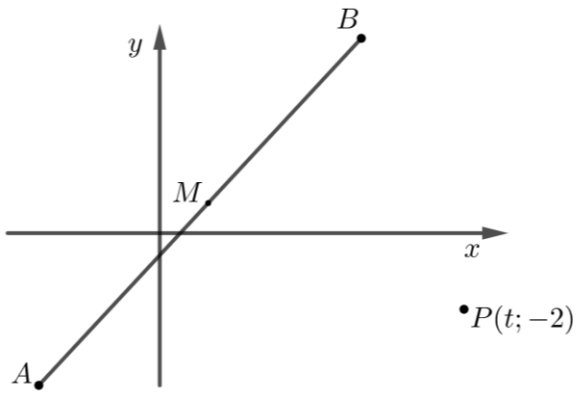
ANALYTICAL GEOMETRY

QUESTION 3

3.1 The points $A(-2\,;k)$, $B(2\,;-1)$ and $C(5\,;-k)$ are colinear. Determine the value of k. (4)

3.2 $A(-3\,;\,-11)$, $B(0\,;\,-2)$ and $C(t\,;10)$ are colinear. $D(9\,;n)$ is a point such that $DB \perp AB$. Determine the values of t and n. (8)

3.3 In the diagram $M(1\,;m)$ is the midpoint of the line AB with $A(-4\,;\,-3)$ and $B(n\,;5)$. $P(t\,;\,-2)$ is a point in the 4th quadrant.

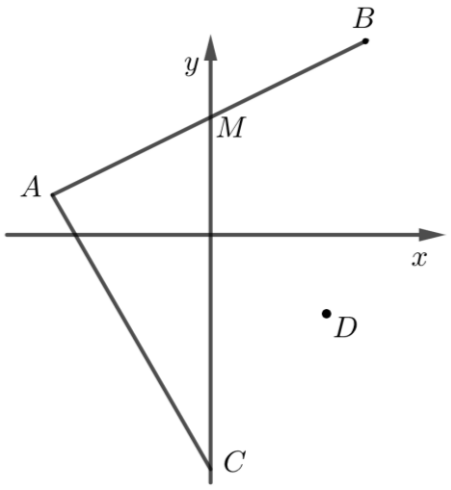


3.3.1 Determine the values of m and n. (3)

3.3.2 If $MP = \sqrt{58}$, determine the value of t. (5)

3.3.3 Determine the radius of the circle centre M and passing through A and B, leaving your answer in simplest surd form. (3)

3.4 In the sketch the line AB has $A(-4\,;m)$ and $B(n\,;5)$. Point C lies on the y-axis such that $AC \perp AB$. Point $D(3\,;k)$ lies in the 4th quadrant.



3.4.1 If $M(0\,;3)$ is the midpoint of AB, determine the values of m and n. (3)

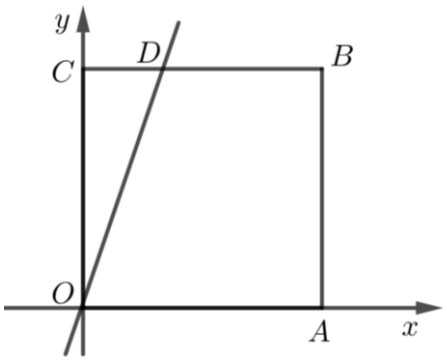
3.4.2 If $n = 4$ and $m = 1$, determine the gradient of AB. (2)

3.4.3 Hence determine the length MC. (5)

3.4.4 Hence calculate the area of $\triangle AMC$. (2)

3.4.5 If $BD = 5\sqrt{2}$, determine the value of k. (5)

3.5 In the diagram, $OCBA$ is a square with side 9 units. OD is the line $\frac{1}{3}y = x$, with D on CB.



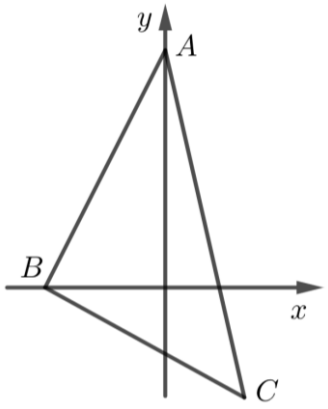
3.5.1 Describe the quadrilateral $ODBA$. (1)

3.5.2 Hence determine the area of $ODBA$. (4)

3.5.3 Determine the coordinates of P, where P lies on OD and $AP \perp OD$. (5)

QUESTION 4

4.1 ΔABC lies in the cartesian plane with $A(0\,;6)$, $B(-3\,;0)$ and $C\left(t\,;-\frac{5}{2}\right)$ as shown.



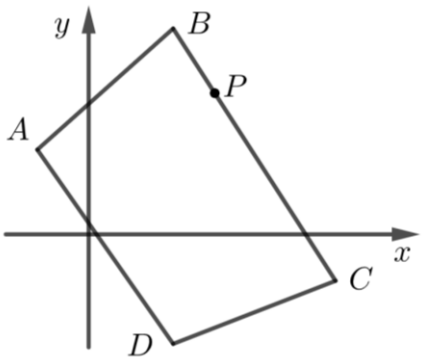
4.1.1 If $A\hat{B}C = 90^{\circ}$, determine the value of t. (4)

4.1.2 $D(x\,;y)$ is a point such that $CD \parallel x$-axis with $x_D > x_C$.

4.1.2.1 Write down a possible coordinate of D. (1)

4.1.2.2 Write down the coordinates of D if $CD = 7$ units. (2)

4.2 Quadrilateral $ABCD$ with $A(-1\,;2)$, $B(2\,;5)$, $C(6\,;-1)$ and $D(2\,;-3)$ is shown. P lies on BP.



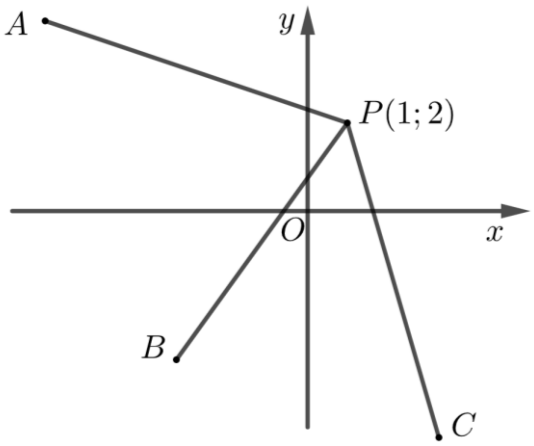
4.2.1 Determine the gradient of CD. (2)

4.2.2 Determine the equation of BC. (4)

4.2.3 Hence determine the coordinates of P such that $APCD$ is a trapezium. (5)

4.2.4 Calculate the area of $ABCD$. (5)

4.3 Points $P(1\,;2)$, $A(-6\,;t)$, $B(n\,;-4)$ and $C(k\,;-2k)$ are shown.



4.3.1 Determine the following:

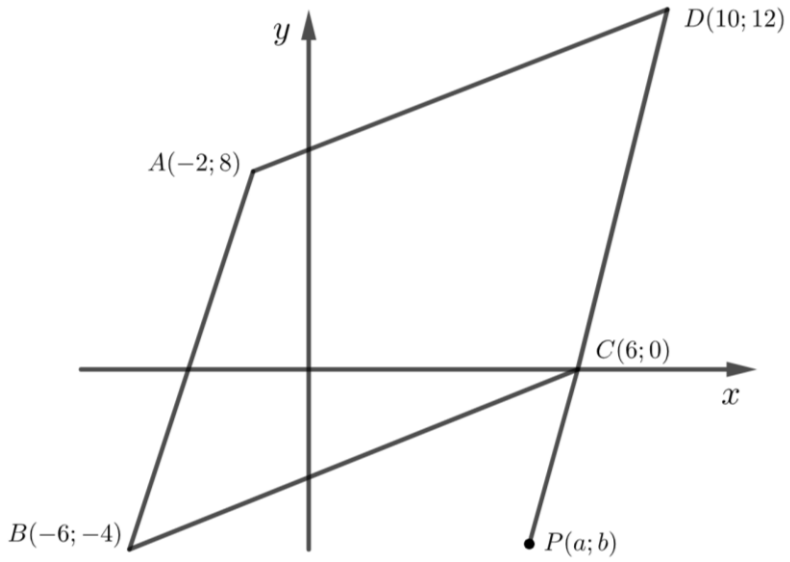
4.3.1.1 t if $PA = \sqrt{58}$ units. (4)

4.3.1.2 n if $m_{PB} = \frac{3}{2}$. (3)

4.3.1.3 k if P, C, and $D(3\ ;\ -6)$ are colinear. (5)

4.3.1.4 The area of $\triangle QOR$ if the perpendicular bisector of PB has an x-intercept at Q and a y-intercept at R. (6)

4.4 Quadrilateral $ABCD$ is shown in the diagram. DC is extended to $P(a\ ;b)$ such that $CP = \frac{1}{2}DC$.



4.4.1 If Q is the intersection of the diagonals of $ABCD$, prove that $A\hat{Q}D = 90°$. (3)

4.4.2 Prove that $AD = BC$. (4)

4.4.3 It is given that $AD \parallel BC$. Describe with <u>full</u> reasons the specific shape of quadrilateral $ABCD$. (3)

4.4.4 Calculate the values of a and b. (4)

4.4.5 E is a point with a smaller x-value than A and $AE \parallel x$-axis.

4.4.5.1 Write down a possible coordinate of E. (1)

4.4.5.2 Write down the coordinates of E if $AE \perp BE$. (2)

4.4.5.3 Write down the coordinates of E if $AE = 12$ units. (2)

TRIGONOMETRY

QUESTION 5

5.1 If $x = 37°$ and $y = 23°$, determine, showing all calculations, whether the following are True or False:

5.1.1 $\cos(x + y) = 0{,}5$ (2) 5.1.2 $\sin(x - y) = sinx - siny$ (2)

5.1.3 $sin^2x + cos^2y = 1$ (2) 5.1.4 $cos2y = 2cos^2y - 1$ (2)

5.1.5 $sin2x = 2sinx.cosx$ (2) 5.1.6 $\tan(x + y) = \frac{\sin(x+y)}{\cos(x+y)}$ (2)

5.2 Simplify the following showing all working and without the use of a calculator:

5.2.1 $2sin30°.cos30°$ (3) 5.2.2 $tan^260° - 3$ (2)

5.2.3 $sec45° + \frac{1}{\sqrt{2}} \times sin90°$ (3) 5.2.4 $\frac{7}{4} - cosec30° + cos^260°$ (3)

5.2.5 $sin30°.cosec30°.cos30°.tan30°$ (4)

5.2.6 $(cot30°)^2 - 2(sin^260° + cos^260°)$ (4)

5.2.7 $\sqrt{3} \times sin90^\circ + \sqrt{2} \times sin45^\circ - 2 \times cos30^\circ$ (4)

5.2.8 $2.sin45^\circ.cos45^\circ + sin^230^\circ - 1$ (3) 5.2.9 $\frac{cot60^\circ}{cosec60^\circ}$ (4)

5.3 Without the use of a calculator and showing all working, determine the value of the acute angle B if: $\frac{1}{\sqrt{3}}sin60^\circ + sinB - tan45^\circ = 0$ (4)

5.4 $T = \frac{3.\sin(x+20^\circ)}{cosy - tan40^\circ}$ Use a calculator to calculate the value of T to 3-decimal places if:

5.4.1 $x = 35,8^\circ$ and $y = 60^\circ$ (2) 5.4.2 $x = 130^\circ$ and $y = x + 20^\circ$ (3)

5.5 Use a calculator to determine the value of the following to 1-decimal place:

5.5.1 $\frac{1}{3}cos(25^\circ + 77^\circ) - sin3^\circ$ (2) 5.5.2 $sec27^\circ \times tan12^\circ + cosec147^\circ$ (3)

5.5.3 $\frac{3}{10} \times tan50^\circ - sin^225^\circ + cos^2(12^\circ + 19^\circ) + cot40^\circ$ (3)

QUESTION 6

6.1 Given that $-13 \times cos\theta - 5 = 0$ and $tan\theta > 0$, without a calculator, use a diagram to determine the value of:

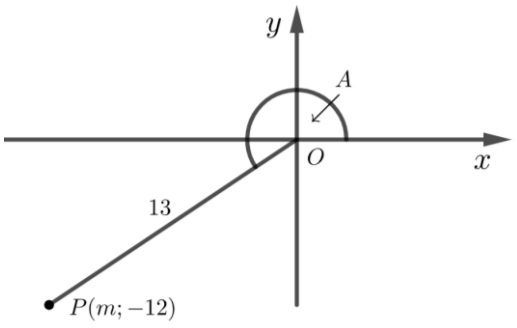
6.1.1 $sin\theta$ (4) 6.1.2 $sec\theta . cot\theta$ (4) 6.1.3 $\frac{169}{144} - cosec^2\theta$ (3)

6.2 Given that $25.sinA + 7 = 0$ and $cosA > 0$, without using a calculator, determine the value of:

6.2.1 $cosA$ (3) 6.2.2 $sin^2A + (cosA)^2$ (3)

6.2.3 $secA \times \frac{1}{cotA}$ (3) 6.2.4 $1 + tanA - \frac{1}{sinA}$ (3)

6.3 In the diagram, $OP = 13$ with $P(m\,;\,-12)$ and $x\hat{O}P = A$.

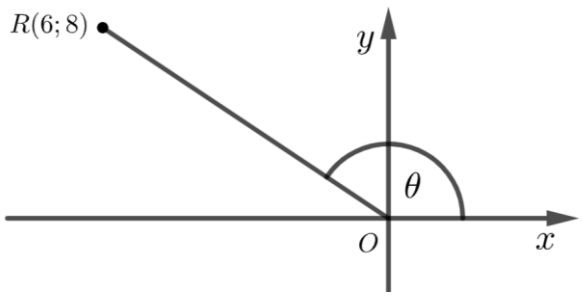


Without using a calculator, determine the value of:

6.3.1 m (2) 6.3.2 $cosA$ (2)

6.3.3 $cosecA + sinA$ (3) 6.3.4 $\frac{1}{cotA} - tanA.secA$ (3)

6.4 In the diagram $R(6\,;8)$ lies on the line OR. $x\hat{O}R = \theta$.

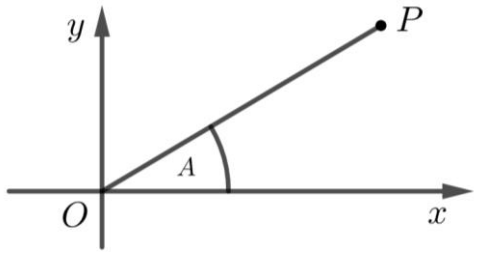


Without using a calculator, determine the value of:

6.4.1 the length OR. (2) 6.4.2 $1 - 2sin\theta \times cos\theta$ (3)

6.4.3 $cos2\theta$ if $cos2\theta = cos^2\theta - sin^2\theta$ (3) 6.4.4 $sec\theta + cot\theta + cosec\theta$ (3)

6.5 In the sketch, OP lies in the cartesian plane with $P(5\,;m)$ and $OP = 13$ units.



6.5.1 Show that $m = 12$. (2)

6.5.2 Without the use of a calculator, determine the value of:

6.5.2.1 $13.sinA - 2$ (2) 6.5.2.2 $cos^2A - 1$ (2) 6.5.2.3 $secA + tanA$ (3)

6.6 If $-tanA = \frac{4}{3}$ and $0^\circ \leq A \leq 180^\circ$, without the use of a calculator, determine the value of the following:

6.6.1 $sinA$ (3) 6.6.2 $secA$ (2) 6.6.3 $\frac{cotA}{1-cosec^2A}$ (4)

6.7 Given that $B \in [0^\circ\,;360^\circ]$, which quadrant(s) will/can angle B lie in if:

6.7.1 $sinB > 0$ and $cosB < 0$? (2)

6.7.2 $cosB > 0$ and $tanB < 0$? (2)

6.7.3 $sinB \times tanB > 0$? (3)

6.8 If $cos\theta = \frac{m}{n}$ where $\theta \in (0^\circ\,;90^\circ)$, use a diagram to determine the value of the following in terms of m and n:

6.8.1 $cosec\theta$ (4) 6.8.2 $m^2.tan^2\theta + n^2.sin^2\theta$ in its simplest form (4)

6.9 If $W = 1 - 4.sin^2\theta$, where $\theta \in [0^\circ\,;90^\circ]$, determine the following:

6.9.1 the maximum value of W (2) 6.9.2 the minimum value of W (2)

6.9.3 the value of θ for which $W = 0$ (4)

QUESTION 7

7.1 Solve the following equations for $x \in [0^\circ ; 90^\circ]$:

7.1.1 $-3.sin2x + 1 = 0$ (2)

7.1.2 $tan\left(\frac{1}{2}x + 40^\circ\right) = 4$ (2)

7.1.3 $\sqrt{5}\,cos(x - 5^\circ) = 2$ (2)

7.1.4 $4.sin\frac{x}{3} = \sqrt{3}$ (3)

7.1.5 $\pi.cosx - 2 = 0{,}5$ (2)

7.1.6 $\frac{\sqrt{3}}{4}\,sin(x + 25^\circ) = \frac{1}{3}$ (3)

7.1.7 $6.cos3x = tan80^\circ$ (3)

7.1.8 $\sqrt{7} \times sin\frac{x}{2} = cos70^\circ + \sqrt{3} \times sin\frac{x}{2}$ (4)

7.1.9 $\frac{cosec2x}{cos50^\circ} = \sqrt{3}$ (3)

7.1.10 $\frac{3}{sec(x-20^\circ)} = \frac{\sqrt{7}}{2}$ (3)

7.2 Solve simultaneously for x and y where $x, y \in [0^\circ ; 90^\circ]$:

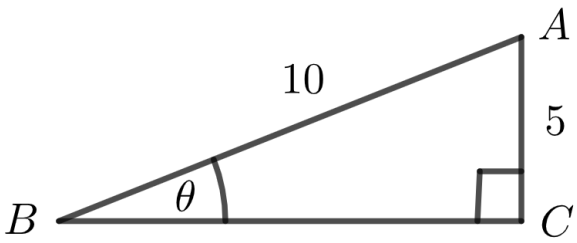
7.2.1 $3.sinx + cosy = \sqrt{2}$ and $2.sinx - cosy = \frac{1}{2}$, (6)

7.2.2 $sin2x - cosy = 0{,}2$ and $sin2x = 0{,}9$ (4)

7.2.3 $tan2x - cos2y = 5$ and $3cos2y - tan2x = -3{,}5$ (6)

QUESTION 8

8.1 Right-angled triangle ABC is shown in the figure.



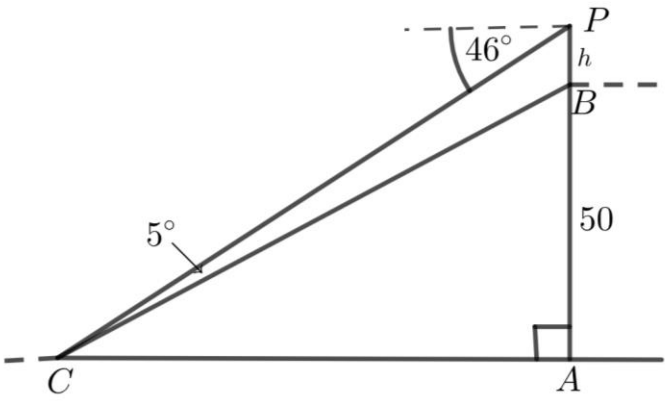
Without the use of a calculator and showing all working:

8.1.1 determine the value of θ. (2)

8.1.2 hence use θ to calculate the length BC in simplest surd form. (3)

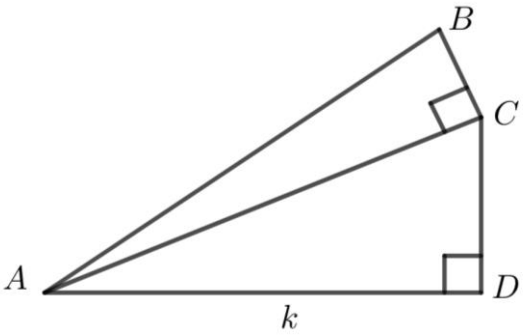
8.1.3 determine BC by applying Pythagoras's Theorem in simplest surd form. (2)

8.2 In the sketch, CA is the horizontal ground level. The side AB of a building is perpendicular to CA with $AB = 50$ metres. PB is a vertical pole height h and $B\hat{C}P = 5^\circ$. The angle of depression of C from P is 46°.



Determine the value of h. (6)

8.3 In the sketch, triangles ABC and ADC are right-angled as shown. $B\hat{A}D = 60^\circ$ and $AD = k$ units.



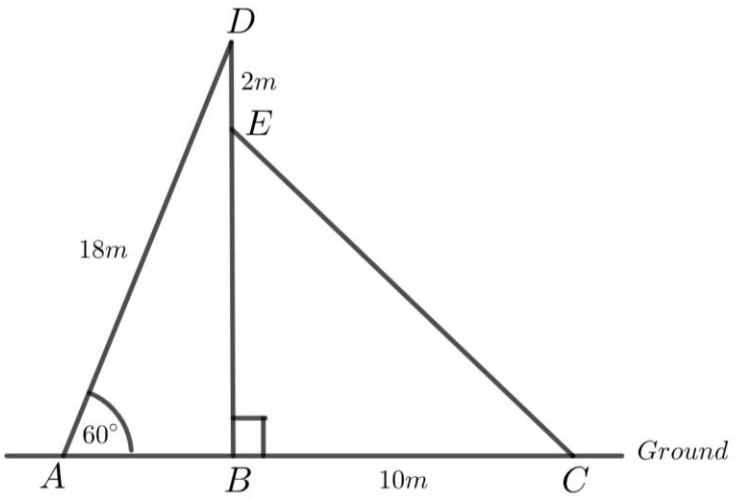
If AC bisects the angle at A, show, without using a calculator, that $BC = \frac{2}{3}k$. (5)

8.4 In the diagram, A, B, C, D, and E all lie in the same vertical plane. ABC is the horizontal ground level. BD is a vertical flagpole which is supported by two steel cables, AD and CE. The angle of elevation of D from A is 60°.
$AD = 18\ metres$; $BC = 10\ metres$; $DE = 2\ metres$

8.4.1 Calculate the height of the flagpole BD, leaving your answer in simplest surd form) (2)

8.4.2 Hence calculate the length of the steel cable CE. (4)

8.4.3 Determine the angle of elevation of E from C, correct to ONE decimal place. (2)



8.5 In Figure 1, ladder AB is 9 metres long with A on the ground and B against a vertical wall WB as shown. The ladder is not securely positioned and it slips down the wall such that $BB_1 = 1{,}02$ metres. The ladder is now in position A_1B_1 as shown in Figure 2. All points lie in the same vertical plane.

Figure 1

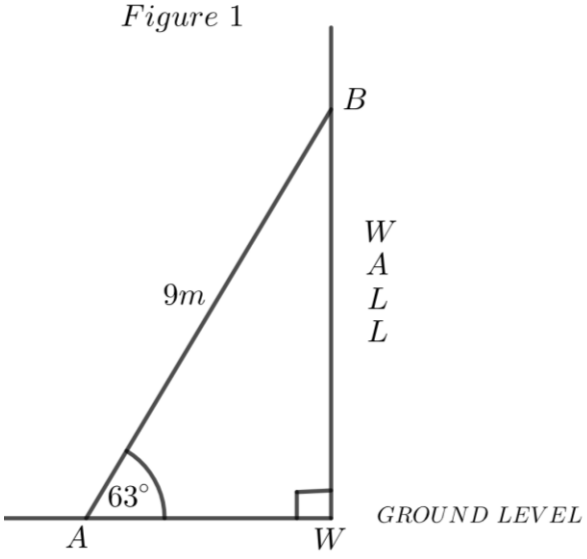
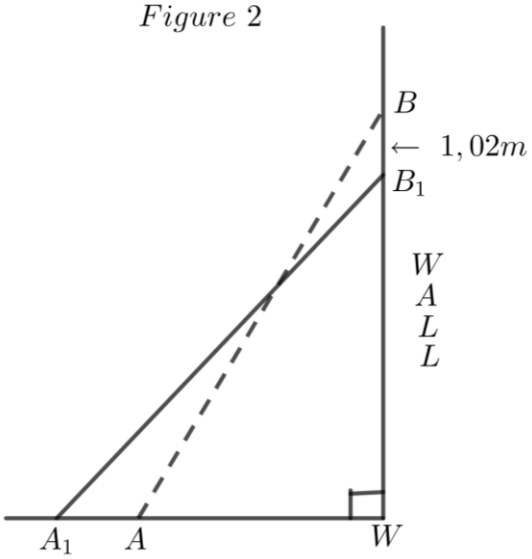


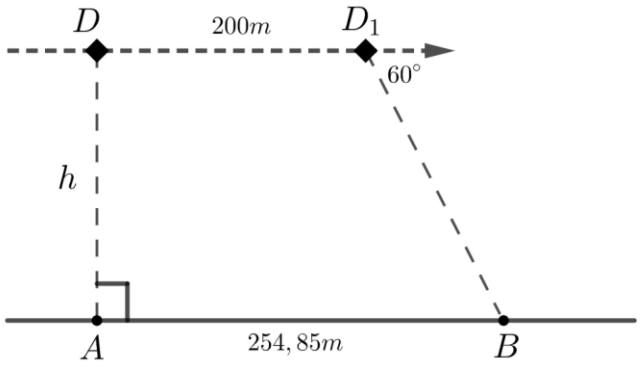
Figure 2



8.5.1 Determine the height of B above W. (Correct to TWO decimal places) (2)

8.5.2 Calculate the distance AA_1. (5)

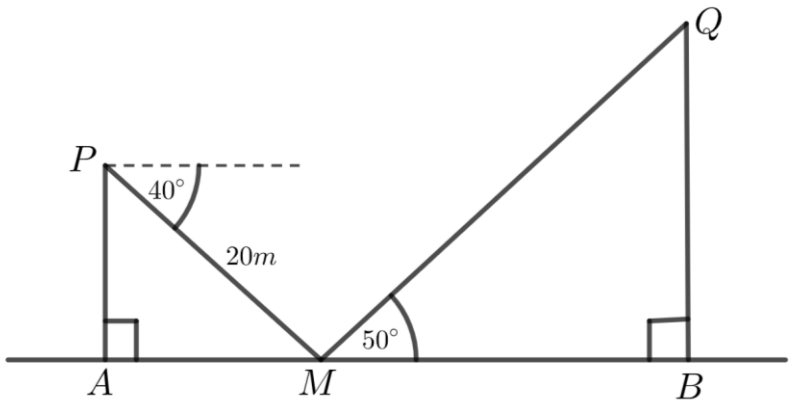
8.6 In the diagram, AB represents the horizontal ground level. A drone, D, is h metres vertically above A. The drone travels 200 metres parallel to ground level AB to D_1. The angle of depression of B from D_1 is 60°. $AB = 254{,}85$ metres.



8.6.1 Determine the value of h. (4)

8.6.2 How long will it take the drone to travel from D to D_2 if it is flying at a constant speed of $15ms^{-1}$ and D_2 is vertically above B? (Answer to the nearest second) (3)
Note: $Distance = Speed \times Time$

8.7 In the figure AMB is the horizontal ground level. AP and BQ are vertical poles with $AP = \frac{1}{2}BQ$. $PM = 20$ metres. A, M, B, P and Q all lie in the same vertical plane. The angle of depression of M from P is 40° and the angle of elevation of Q from M is 50°.



8.7.1 Determine the distance AB. (7)

8.7.2 Determine the distance PQ. (6)

QUESTION 9

9.1 $f(x) = sinx$ and $g(x) = 2cosx - 1$, $x \in [0^\circ ; 360^\circ]$

9.1.1 On the same set of axes, sketch neat graphs of the two functions, clearly showing all axes intercepts and turning points. Label the two points of intersection of the graphs as A and B with $x_A < x_B$. (8)

9.1.2 Write down the:

9.1.2.1 amplitude of f and the amplitude of g. (2) 9.1.2.2 range of g. (1)

9.1.3 Write down the coordinates of B. (2)

9.1.4 Given $A(36.87^\circ ; 0{,}60)$, for which values of x in the domain is $g(x) > f(x)$? (4)

9.1.5 $h(x) = f(x) + k$. For which values of k will h not have any x-intercepts? (2)

9.1.6 Excluding any turning points, for which values of x is $f(x)$ decreasing? (2)

9.2 $f(x) = \frac{1}{2}cosx + 1$ and $g(x) = tanx - 1$, $x \in [0^\circ ; 360^\circ]$

9.2.1 On the same set of axes, sketch neat graphs of the two functions, clearly showing all axes intercepts, turning points and asymptotes. Label the two points of intersection of the graphs as P and Q with $x_P < x_Q$. (8)

9.2.2 Write down the:

9.2.2.1 period of g (1) 9.2.2.2 range of f (1) 9.2.2.3 amplitude of f (1)

9.2.3 Given that $x_P = 65{,}62^\circ$ and $x_Q = 240{,}29^\circ$, for which values of x in the given domain is $g(x) \leq f(x)$? (6)

9.2.4 $h(x) = f(x) + m$. For which values of m will h have at least two x-intercepts in the given domain? (2)

9.3 The graph of $f(x) = a.sinx + b$ is shown in the sketch for $x \in [0^\circ ; 360^\circ]$.
The y-intercept of f is $\left(0^\circ ; -\frac{1}{2}\right)$.

9.3.1 Write down the values of a and b. (2)

9.3.2 Write down the coordinates of P, a turning point of f. (1)

9.3.3 What is the:

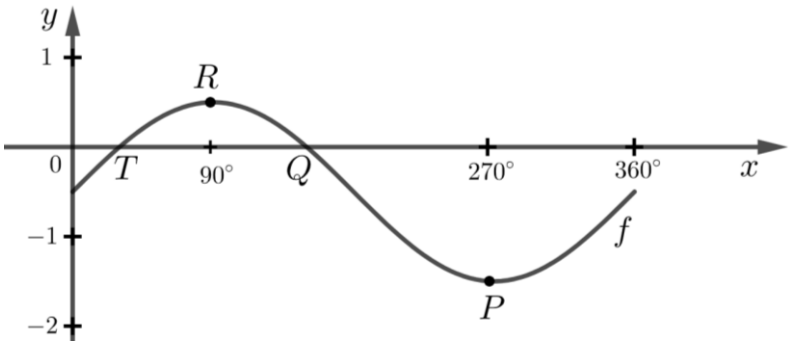
9.3.3.1 amplitude of f ? (1) 9.3.3.2 the range of f ? (1)

9.3.4 Solve for θ if $sin\theta - \frac{1}{2} = 0$, $0^\circ < \theta < 180^\circ$ (2)

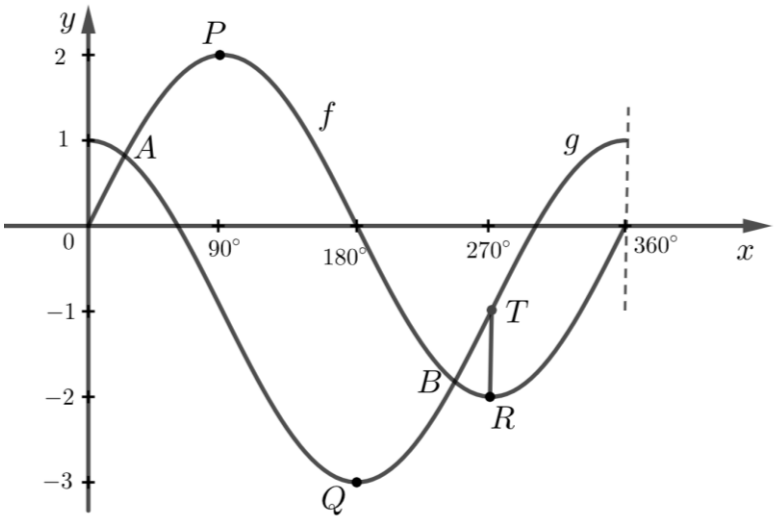
9.3.5 Hence or otherwise give the coordinates of the point(s) from the graph for which $f(x) = 0$. (2)

9.3.6 $g(x) = f(x) + k$. For which value(s) of k will the graph of g be:

9.3.6.1 always positive? (1) 9.3.6.2 always negative? (1)



9.4 In the diagram the graphs of $f(x) = m.sinx$ and $g(x) = 2.cosx + n$ are shown for $x \in [0^\circ ; 360^\circ]$. $Q(180^\circ ; -3)$ is a turning point on g. P and R are turning points on f. The two graphs intersect at $A(24{,}30^\circ ; 0{,}82)$ and $B(245{,}70^\circ ; -1{,}82)$.



9.4.1 Write down the values of m and n. (2)

9.4.2 Write down the coordinates of P and Q. (2)

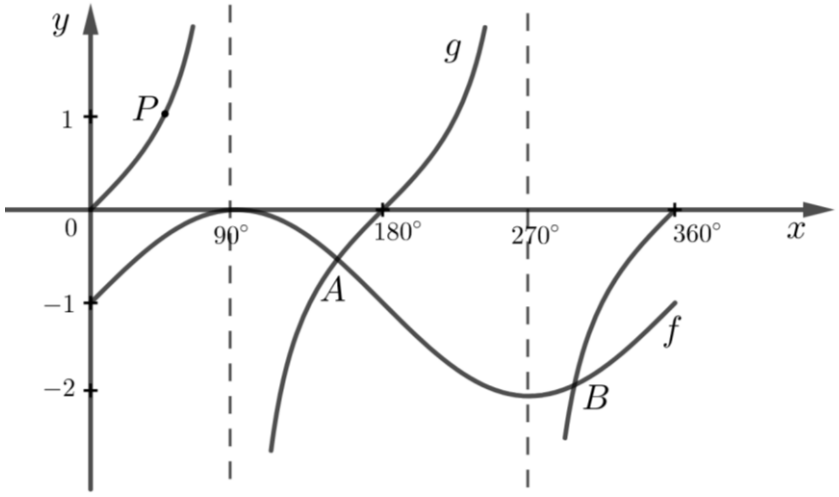
9.4.3 $TR \parallel y$-axis with T on g. Determine the length TR. (2)

9.4.4 Determine the coordinates where $g(x) = 0$. (3)

9.4.5 For which values of x is:

9.4.5.1 $f(x) \leq 0$? (2) 9.4.5.2 $g(x) > f(x)$? (2) 9.4.5.3 $f(x).g(x) > 0$? (4)

9.5 In the sketch, $f(x) = sinx + m$ and $g(x) = tanx$ are shown for $x \in [0° ; 360°]$. $P(n ; 1)$ lies on g. A and B are the points of intersection of the two graphs with x_A=152,04° and $x_B = 297{,}97°$.



9.5.1 Write down the value of m. (1)

9.5.2 What is the range of $f(x)$? (1)

9.5.3 Write down the value of n. (1)

9.5.4 For which values of x is:

9.5.4.1 $f(x) > g(x)$? (4)

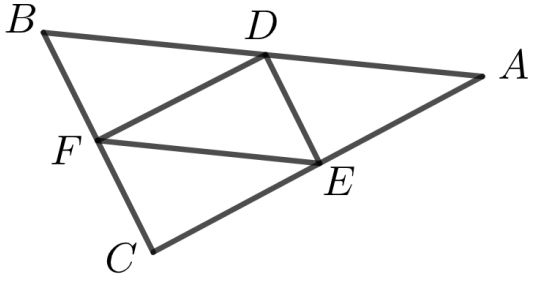
9.5.4.2 $f(x).g(x) = 0$? (3)

9.5.5 $h(x) = f(x) + k$. For which value(s) of k will h have exactly one x-intercept? (2)

ECLUDIEN GEOMETRY

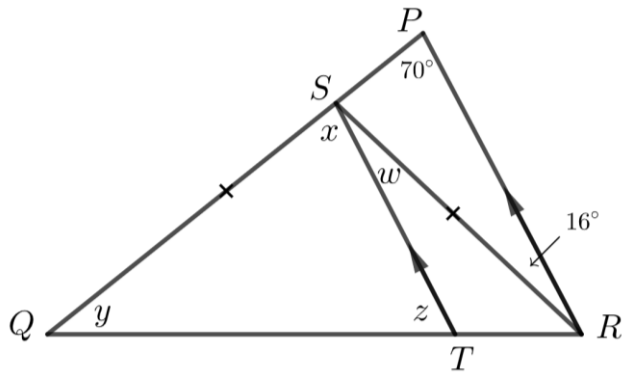
QUESTION 10

10.1 Triangle DEF is inscribed within $\triangle ABC$ such that D, E, and F are the midpoints of AB, AC, and BC respectively. $DF = 7 - x$; $EF = 3x - 5$; $DE = 8 - 2x$



Determine, with reasons, the perimeter of $\triangle ABC$. (4)

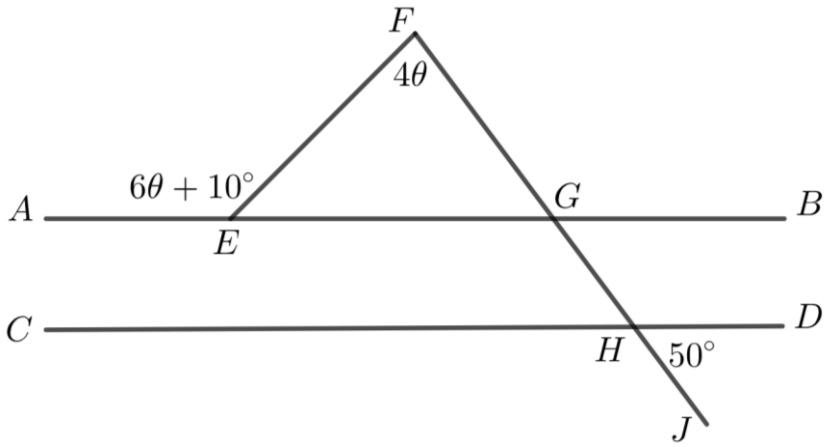
10.2 In the diagram $ST \parallel PR$ and $SR = SQ$. $\hat{P} = 70°$ and $P\hat{R}S = 16°$.



Determine, with reasons, the size of angles w, x, y, and z. (7)

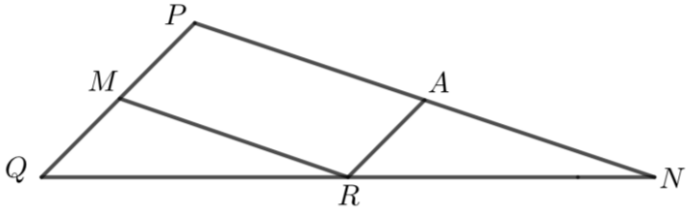
10.3 In the given figure, $AEGB$, $FGHJ$, and CHD are straight lines. $FE = FG$

$A\hat{E}F = 6\theta + 10°$; $\hat{F} = 4\theta$; $J\hat{H}D = 50°$



Prove <u>with full reasoning</u> that $AB \parallel CD$. (7)

10.4 In the figure, $PARM$ is a parallelogram and $QM = PM$.



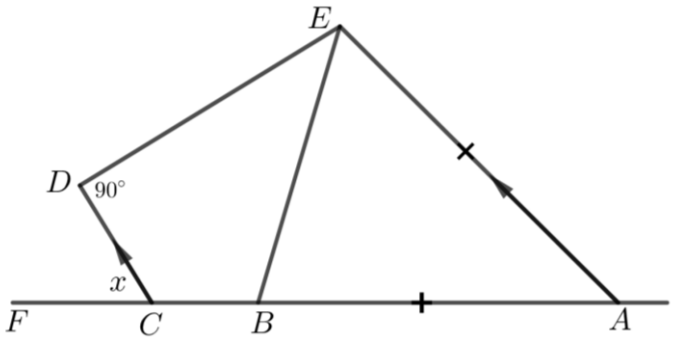
10.4.1 Prove, with reasons, that $\triangle QMR \;|||\; \triangle QPN$. (4)

10.4.2 Prove, with reasons, that $\triangle NRA \equiv \triangle RQM$. (4)

10.4.3 If the perpendicular distance from P to MR is $6cm$ and the area of the parallelogram is $72cm^2$, determine the length of AN. (4)

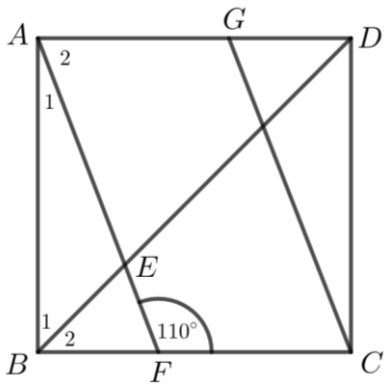
QUESTION 11

11.1 In the figure $AE \parallel CD$, $C\hat{D}E = 90°$, $AE = AB$, and $D\hat{C}F = x$



Determine with full reasons, the value of $B\hat{E}D$ in terms of x. (5)

11.2 In the figure $ABCD$ is a square. AF intersects diagonal BD at E. $A\hat{F}C$=110°.



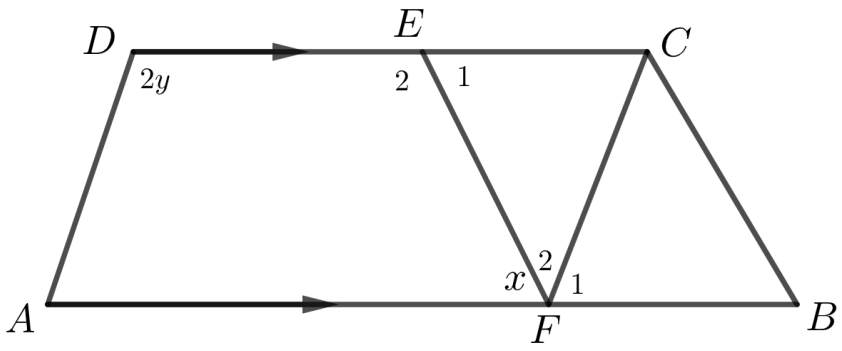
With full reasoning:

11.2.1 determine the value of $A\hat{E}D$. (5)

11.2.2 if $DG = BF$, prove that $AF = CG$. (5)

11.2.3 prove that $AGCF$ is a parallelogram. (4)

11.3 Trapezium $ABCD$ with $AB \parallel CD$ is shown. F lies on AB and E lies on CD such that $FC = FE = FB$. $A\hat{D}C = 2y$ and $A\hat{F}E = x$.



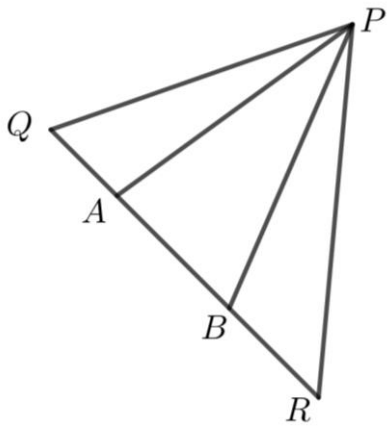
With full reasons:

11.3.1 write down the value of $B\hat{A}D$ in terms of y. (2)

11.3.2 prove that $A\hat{B}C = B\hat{C}F = 90° - \frac{x}{2}$ (6)

11.3.3 hence determine the value of x in terms of y if $ABCD$ is an isosceles trapezium. (3)

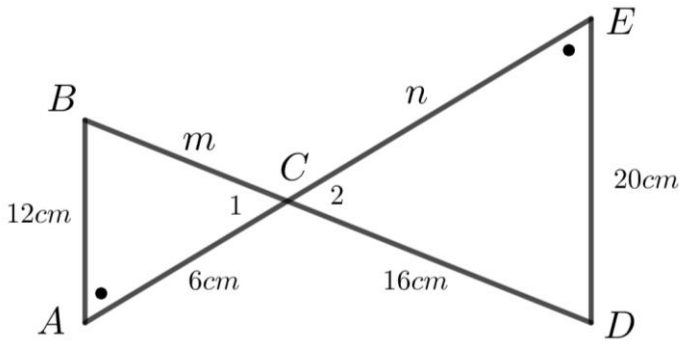
11.4 In the figure, A and B lie on side QR of ΔPQR such that $QA = RB$. $PQ = PR$.



11.4.1 Prove, with reasons, that $\Delta APQ \equiv \Delta BPR$. (4)

11.4.2 Prove, with reasons, that $P\hat{A}B = P\hat{B}A$. (3)

11.5 In the figure, $\hat{A} = \hat{E}$.



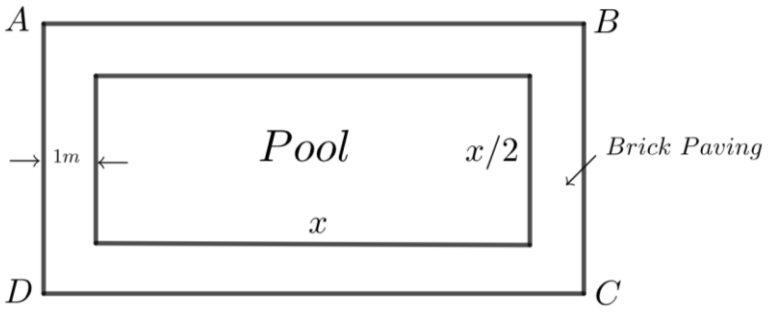
11.5.1 Prove that $\Delta ABC \;|||\; \Delta EDC$. (4)

11.5.2 Hence determine the values of m and n. (4)

MEASUREMENT

QUESTION 12

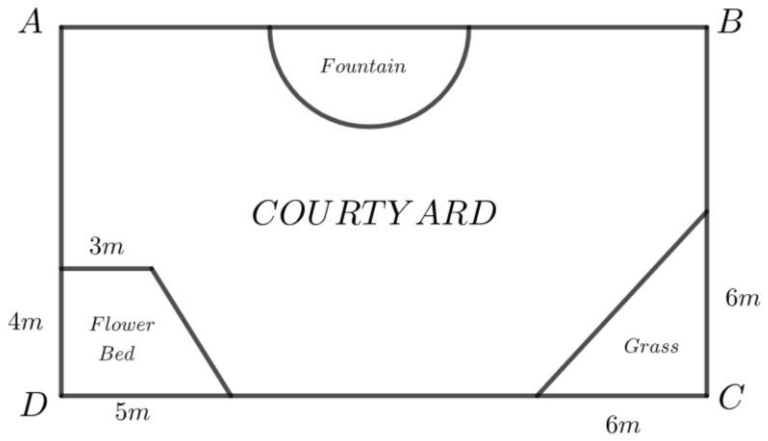
12.1 A rectangular swimming pool has dimensions x metres and $\frac{x}{2}$ metres as shown. The pool is surrounded by brick paving 1 metre wide.



If the area of the rectangle $ABCD$ is $40m^2$, determine the dimensions of the swimming pool (5)

12.2 A school wishes to pave a rectangular courtyard $ABCD$. The area to be paved excludes a semi-circle fountain, flower bed, and grass area as shown.

$AB = 20m, \quad BC = 15m$. The diameter of the fountain is $\frac{12}{\sqrt{\pi}}$ metres.



12.2.1 Determine the area of the courtyard to be paved. (8)

12.2.2 If the pavers cost R90/m^2, what is the cost of the required pavers? (2)

12.3 A closed cylindrical cardboard container holds exactly 8 table tennis balls each of radius $20mm$. The balls are each tangent to the ball(s) on either side and to applicable surfaces of the container as shown.



12.3.1 Determine the total surface area of the cardboard container. Give your answer to the nearest square centimetre. (3)

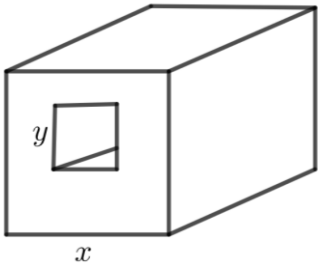
12.3.2 Determine the volume of air between the balls and the inside of the container. Give your answer to the nearest cubic centimetre. (5)

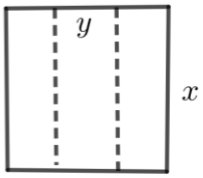
Useful formulae: $SA = 2\pi r^2 + 2\pi rh$ and $V_s = \frac{4}{3}\pi r^3$

12.4 A solid cube with side x metres has a rectangular prism with square base y metres uniformly cut out of it as shown. The cut out is perpendicular to the two faces of the cube as shown. The volume of the remaining solid is $1m^3$.

12.4.1 Express y in terms of x. (3)

12.4.2 Hence or otherwise, if $x = 1{,}1$ metres, determine the value of y, giving your answer in centimetres to 2 decimal places. (3)





Top View

QUESTION 13

13.1 A solid cuboid has dimensions x, $2x$, and $5x$ as shown in Figure 1. The cuboid is cut diagonally into two identical wedges A and B as shown by the dotted line as shown in Figure 2. In Figure 3, wedge B is cut into two pieces C and D as shown by the perpendicular dotted line.

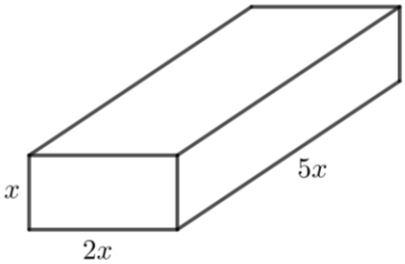


Figure 1

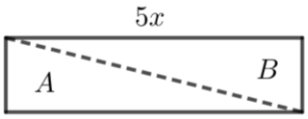


Figure 2

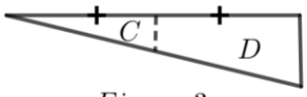


Figure 3

13.1.1 Show that the surface area of shape A may be given by $\left(17 + 2\sqrt{26}\right)x^2$. (6)

13.1.2 Show that the volume of shape D may be given by $\frac{15}{4}x^3$. (5)

13.2 A rectangular prism has sides $3a$, $2a$, and a centimetres. The total surface area of the prism is $550cm^2$.

13.2.1 Determine the value of a. (4)

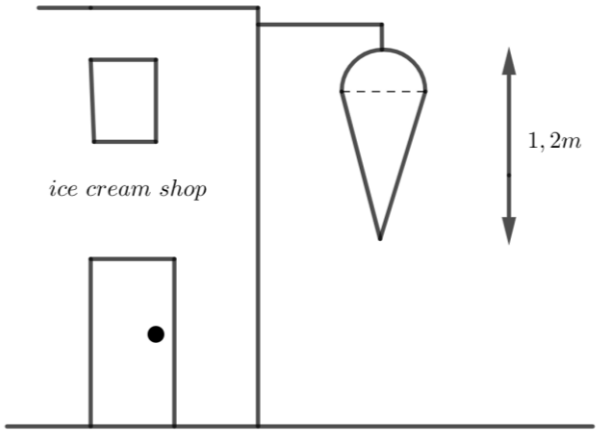
13.2.2 If all sides are multiplied by a factor of k the total surface area reduces to $137{,}5cm^2$. Determine the value of k. (4)

13.3 A rectangular swimming pool has dimensions $20m \times 15m$. The pool is $3m$ deep to the top.

13.3.1 If the water is $200mm$ from the top of the pool, determine the volume of water in the pool. Give your answer in kilolitres. (3)

13.3.2 If every person who climbed into the pool displaced a volume of 150 litres, how many people could be in the pool before it overflowed? (4)

13.4 An ice cream shop displays a large metal ice cream advertising its business. The shape is hollow and made up of a cone and a hemisphere as shown. The vertical height of the display from top to bottom is 1,2 metres and the radius of the hemisphere is 0,3 metres.

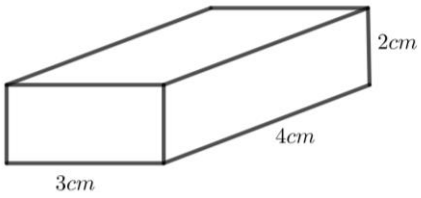


13.4.1 Show that the total volume of the metal ice cream is $\frac{9}{200}\pi$ cubic metres. (4)

13.4.2 Calculate the total surface area of the metal ice cream. (4)

Useful formulae: $V_S = \frac{4}{3}\pi r^3$, $SA_S = 4\pi r^2$, $V_C = \frac{1}{3}\pi r^2 h$, $SA_C(open) = \pi rs$

13.5 A cylindrical jug has a diameter of 0,12m and a height of 30cm. An ice tray has a number of identical compartments each a rectangular prism in shape and measuring $3cm \times 4cm \times 2cm$.



Typical Ice Compartment

13.5.1 If the jug is completely full of water, how may complete ice blocks can it make from a single jug of water where each ice compartment is filled to the top? (5)

13.5.2 If the jug was half-filled with water and a solid glass cylinder of radius 3cm and length 5cm was dropped into it, determine how much the water rises in the jug. (6)